Application Flow

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

1. Entry points

User may configure a Sky Chart, Scientific Diagram or Artistic Composite manually; optionally enter a natural-language prompt. Manual controls are canonical and always available. Suggested local UI progression: Swagger/CLI first; a lightweight frontend only after the rendering pipeline works.

2. Flow A — individual constellation

Choose Orion -> load catalog/line definitions -> select projection and preset -> validate -> run Python renderer -> inspect scientific checks -> export SVG/PNG -> persist manifest -> link to atlas chapter/figure.

3. Flow B — Summer Triangle

Choose Asterism/SummerTriangle -> resolve Vega, Deneb, Altair by stable star IDs -> include Lyra/Cygnus/Aquila lines if requested -> calculate framing -> render distinct asterism overlay -> verify connectivity -> export.

4. Flow C — southern horizon

Choose HorizonView -> set observer coordinates and UTC date/time -> choose south azimuth center and field of view -> transform catalog stars to AltAz -> clip below horizon -> draw azimuth/altitude references and labels -> validate against trusted planetarium reference -> export.

5. Flow D — Polaris mathematical illustration

Choose ScientificDiagram/StarAltitude -> enter observer location and optional UTC time -> choose approximate teaching mode or exact observed mode -> compute NCP altitude and optionally Polaris AltAz -> render ground/horizon, north direction, line of sight, angle arc, zenith and explanatory caption -> export. If exact mode lacks time, prompt user for date/time. If approximate mode, label 42.7 degrees as an approximation for Prishtina rather than an exact instantaneous Polaris value.

6. Flow E — natural-language illustration

Prompt -> LLM returns strictly typed intent and parameters -> validate enum, location, time and template -> show resolved configuration for review -> invoke deterministic scientific renderer -> preview -> optional user correction -> export. Unknown diagram types or conflicting directions must be rejected/clarified, not guessed. Model-generated text is not accepted as numeric proof.

7. Flow F — scenic composite

Generate or choose star-free background -> verify license/asset metadata -> render chart independently -> composite background beneath locked chart layer -> compare scientific geometry hash -> export marked illustrative composite.

8. Job lifecycle and failure recovery

Job statuses: Queued, Running, Completed, Failed, Cancelled. Save validation errors, structured renderer logs, output paths and hashes. Retry only safe idempotent operations. A failed AI call falls back to manual entry; missing catalog or unsupported projection stops scientific render; invalid geometry fails validation rather than producing a plausible-looking image. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

Flow F - planetary conjunction at sunset

1. Select city or coordinates, calendar date, Venus/Jupiter, West, and 30 minutes after sunset. 2. Resolve local date and time zone. 3. Compute local sunset using explicit horizon/refraction convention; if no sunset, return an informative error. 4. Calculate observation UTC, topocentric ephemerides and separation. 5. Check both bodies are above horizon and describe whether they are actually close. 6. Render twilight background, horizon, planet positions, labels and optional angle arc. 7. Save manifest with ephemeris/kernel, calculation settings and outputs.

Flow G - date-range event discovery

User asks for the closest Moon-Venus approach during a month from Sydney. Resolve a bounded date range, sample apparent separation, refine local minima, evaluate local twilight/altitude constraints, rank observable candidate instants and request confirmation before exporting an illustration.

Flow H - compare locations

For one UTC instant or separate local sunset-relative instants, render side-by-side sky scenes from Rome and Prishtina. Explicitly label which time alignment was used. Recalculate each view independently; do not transform pixels from a single-location render.

Failure flows

Unsupported date by ephemeris, missing time zone, DST gap/overlap, invalid coordinates, missing sunset/sunrise, below-horizon target, insufficient angular closeness, unavailable event within search range, and optional AI interpretation uncertainty all return actionable errors/warnings.

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.